

1. The sum of all the roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is
[2023 (06 Apr Shift 1)]
(1) $9 - \sqrt{3}$
(2) $9 + \sqrt{3}$
(3) $11 - \sqrt{3}$
(4) $11 + \sqrt{3}$
2. Let α, β, γ be the three roots of the equation $x^3 + bx + c = 0$ if $\beta\gamma = 1 = -\alpha$ then $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ is equal to
[2023 (08 Apr Shift 1)]
(1) $\frac{155}{8}$
(2) 21
(3) $\frac{169}{8}$
(4) 19
3. Let m and n be the numbers of real roots of the quadratic equations $x^2 - 12x + [x] + 31 = 0$ and $x^2 - 5|x + 2| - 4 = 0$ respectively, where $[x]$ denotes the greatest integer $\leq x$. Then $m^2 + mn + n^2$ is equal to
[2023 (08 Apr Shift 2)]
4. If a and b are the roots of the equation $x^2 - 7x - 1 = 0$, then the value of $\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$ is equal to
[2023 (11 Apr Shift 1)]
5. The number of points, where the curve $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$ cuts x -axis, is equal to.....
[2023 (11 Apr Shift 2)]
6. Let α, β be the roots of the quadratic equation $x^2 + \sqrt{6}x + 3 = 0$. Then $\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$ is equal to
[2023 (12 Apr Shift 1)]
(1) 81
(2) 9
(3) 72
(4) 729
7. Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$ Then $\alpha^{14} + \beta^{14}$ is equal to
[2023 (13 Apr Shift 2)]
(1) -64
(2) $-64\sqrt{2}$
(3) -128
(4) $-128\sqrt{2}$
8. The number of real roots of the equation $x|x| - 5|x + 2| + 6 = 0$, is
[2023 (15 Apr Shift 1)]
(1) 5
(2) 4
(3) 6
(4) 3

ANSWER KEYS

1. (2) 2. (4) 3. (9) 4. (51) 5. (2) 6. (1) 7. (3) 8. (4)

1. (2)

Given,

$$|x^2 - 8x + 15| - 2x + 7 = 0$$

Now taking Case:I when,

$$x^2 - 8x + 15 \geq 0 \Rightarrow (x - 3)(x - 5) \geq 0$$

$$\Rightarrow x \leq 3 \text{ OR } x \geq 5$$

$$|x^2 - 8x + 15| - 2x + 7 = 0$$

$$\Rightarrow x^2 - 8x + 15 - 2x + 7 = 0$$

$$\Rightarrow x^2 - 10x + 22 = 0$$

$$\Rightarrow x = 5 + \sqrt{3}, 5 - \sqrt{3} \text{ (rejected)}$$

$$\text{So, } \alpha = 5 + \sqrt{3}$$

Now taking Case:II when,

$$|x^2 - 8x + 15| - 2x + 7 = 0$$

$$\Rightarrow x^2 - 8x + 15 < 0 \Rightarrow 3 < x < 5$$

$$\Rightarrow x^2 - 8x + 15 + 2x - 7 = 0$$

$$\Rightarrow x^2 - 6x + 8 = 0 \Rightarrow x = 4, 2 \text{ (rejected)}$$

$$\text{So, } \gamma = 4$$

$$\text{Hence, } \alpha + \gamma = 9 + \sqrt{3}$$

2. (4)

Given that roots of $x^3 + bx + c = 0$ are α, β, γ

Also given that

$$\beta\gamma = 1 = -\alpha$$

$$\Rightarrow \alpha = -1 \dots (i)$$

Now let us apply the relation between the roots of the cubic equation with the coefficients.

$$\Rightarrow \alpha + \beta + \gamma = 0 \dots (ii)$$

$$\Rightarrow \alpha\beta\gamma = -c \dots (iii)$$

$$\Rightarrow (-1)(1) = -c$$

$$\therefore c = 1 \dots (iv)$$

On substituting the value of $\alpha = -1$ in eq (ii) we get,

$$\beta + \gamma = 1 \dots (v)$$

$$\therefore \alpha\beta + \beta\gamma + \gamma\alpha = b$$

$$\Rightarrow \alpha(\beta + \gamma) + \beta\gamma = b$$

$$\therefore b = 0 \dots (vi)$$

Hence, equation will be,

$$x^3 + 1 = 0$$

Whose roots will be, $-1, -\omega, -\omega^2$

We know that $1 + \omega + \omega^2 = 0$

$$\Rightarrow \alpha + \beta + \gamma = 0$$

$$\Rightarrow -1 + \beta + \gamma = 0$$

$$\therefore \beta = -\omega, \gamma = -\omega^2 \dots (vii)$$

$$\Rightarrow \beta^3 = -\omega^3 = -1 \text{ and } \Rightarrow \gamma^3 = -\omega^6 = -1$$

Hence the value of

$$b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3 \text{ is}$$

$$= 0 + 2 + 3 + 6 + 8$$

$$= 19$$

Hence this is the correct option.

3. (9)

Given,

$$x^2 - 12x + [x] + 31 = 0$$

$$\Rightarrow x^2 - 12x + 31 = -[x]$$

$$\geq -5$$

Now from above equation we say that, it could have its solution in $[5, 6)$ but it does not exist as at $x = 5$ as LHS = 1,

So no solution, hence $m = 0$

Now solving,

$$x^2 - 5|x + 2| - 4 = 0$$

Taking Case 1 when $x \geq -2$ we get,

$$x^2 - 5(x + 2) - 4 = 0$$

$$\Rightarrow x^2 - 5x - 14 = 0 \Rightarrow x = 7, -2$$

Now taking Case 2 when $x < -2$ we get,

$$x^2 + 5x + 10 - 4 = 0$$

$$\Rightarrow x = -2, -3$$

So, total 3 solution i.e., $x = -3, -2, 7$

Hence, $n = 3$

$$\text{So, the value of } m^2 + mn + n^2 = 0 + 0 + 3^2 = 9$$

4. (51)

Since, a and b are roots of $x^2 - 7x - 1 = 0$, so

$$a^2 - 7a - 1 = 0$$

$$\Rightarrow a^2 - 1 = 7a$$

$$\Rightarrow a^4 + 1 - 2a^2 = 49a^2$$

$$\Rightarrow a^4 + 1 = 51a^2 \dots (1)$$

Similarly,

$$b^4 + 1 = 51b^2 \dots (2)$$

Now,

$$\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$$

$$= \frac{a^{17}(a^4 + 1) + b^{17}(b^4 + 1)}{a^{19} + b^{19}}$$

$$= \frac{51a^{19} + 51b^{19}}{a^{19} + b^{19}}$$

$$= 51$$

5. (2)

Given,

$$e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1 = 0$$

$$\Rightarrow e^{4x} - e^{2x} - 3 - \frac{1}{e^{2x}} + \frac{1}{e^{4x}} = 0$$

$$\Rightarrow e^{4x} + \frac{1}{e^{4x}} + 2 - \left(e^{2x} + \frac{1}{e^{2x}}\right) = 5$$

$$\Rightarrow \left(e^{2x} + \frac{1}{e^{2x}}\right)^2 - \left(e^{2x} + \frac{1}{e^{2x}}\right) = 3$$

$$\text{Now let } \left(e^{2x} + \frac{1}{e^{2x}}\right) = t$$

So, the equation becomes

$$t^2 - t - 5 = 0$$

$$\Rightarrow t = \frac{1 + \sqrt{21}}{2}, \text{ ignoring negative sign as exponential function are positive,}$$

$$\text{Now } e^{2x} + \frac{1}{e^{2x}} = \frac{1 + \sqrt{21}}{2}$$

$$\Rightarrow e^{4x} - \frac{1 + \sqrt{21}}{2}e^{2x} + 1 = 0 \text{ which is quadratic equation in } e^{2x} \text{ with upward parabola,}$$

$$\text{Now by } A.M \geq G.M \text{ we get, } e^{2x} + \frac{1}{e^{2x}} \geq 2$$

$$\text{So, } y = \frac{1 + \sqrt{21}}{2} > 2, \text{ hence it will cut at two distinct point,}$$

Hence, there will be two solution.

6. (1)

To find the value of $\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$,

Let $a_n = \alpha^n + \beta^n$

Hence,

$$\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}} = \frac{a_{23} + a_{14}}{a_{15} + a_{10}}$$

Now, $x^2 + \sqrt{6}x + 3 = 0$ has roots α & β

$$\text{So, } x = \frac{-\sqrt{6} \pm \sqrt{\sqrt{6}^2 - 4 \times 3}}{2}$$

$$\Rightarrow x = \sqrt{6} \left(\frac{-1 \pm i}{2} \right)$$

$$\Rightarrow x = \sqrt{3} \left(\frac{-1 \pm i}{\sqrt{2}} \right)$$

$$\text{Hence, } \alpha = \sqrt{3} e^{i\frac{3\pi}{4}} \text{ and } \beta = \sqrt{3} e^{i\frac{5\pi}{4}}$$

Now, solving $\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$

$$= \left[\frac{(\sqrt{3})^{23} \left(e^{i23 \times \frac{3\pi}{4}} + e^{i23 \times \frac{5\pi}{4}} \right) + (\sqrt{3})^{14} \left(e^{i14 \times \frac{3\pi}{4}} + e^{i14 \times \frac{5\pi}{4}} \right)}{(\sqrt{3})^{15} \left(e^{i15 \times \frac{3\pi}{4}} + e^{i15 \times \frac{5\pi}{4}} \right) + (\sqrt{3})^{10} \left(e^{i10 \times \frac{3\pi}{4}} + e^{i10 \times \frac{5\pi}{4}} \right)} \right]$$

$$= (\sqrt{3})^4 \left[\frac{(\sqrt{3})^9 \left(e^{i23 \times \frac{3\pi}{4}} + e^{i23 \times \frac{5\pi}{4}} \right) + \left(e^{i14 \times \frac{3\pi}{4}} + e^{i14 \times \frac{5\pi}{4}} \right)}{(\sqrt{3})^5 \left(e^{i15 \times \frac{3\pi}{4}} + e^{i15 \times \frac{5\pi}{4}} \right) + \left(e^{i10 \times \frac{3\pi}{4}} + e^{i10 \times \frac{5\pi}{4}} \right)} \right]$$

$$= 9 \times \left(\frac{(\sqrt{3})^9 \left(\frac{1+i-1+i}{\sqrt{2}} \right) + 0}{(\sqrt{3})^5 \left(\frac{1+i-1+i}{\sqrt{2}} \right) + 0} \right) \left\{ \text{as } e^{i\frac{21\pi}{2}} + e^{i\frac{35\pi}{2}} = 0 + i \sin \frac{21\pi}{2} + 0 + i \sin \frac{35\pi}{2} = i - i = 0 \right\}$$

$$= 9 \times \left(\frac{(\sqrt{3})^9 \left(\frac{2i}{\sqrt{2}} \right)}{(\sqrt{3})^5 \left(\frac{2i}{\sqrt{2}} \right)} \right)$$

$$= 9 \times (\sqrt{3})^4$$

$$= 81$$

7. (3)

Given,

$$x^2 - \sqrt{2}x + 2 = 0$$

$$\Rightarrow x = \frac{\sqrt{2} \pm \sqrt{\sqrt{2}^2 - 4 \times 2}}{2}$$

$$\Rightarrow x = \sqrt{2} \left(\frac{1 \pm \sqrt{3}i}{2} \right)$$

$$\Rightarrow x = \sqrt{2} \left(\frac{1}{2} \pm \frac{\sqrt{3}i}{2} \right)$$

$$\Rightarrow x = \sqrt{2} \left(\cos \frac{\pi}{3} \pm i \sin \frac{\pi}{3} \right)$$

$$\text{So, } \alpha = \sqrt{2} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = \sqrt{2} e^{i\frac{\pi}{3}}$$

$$\text{And } \beta = \sqrt{2} \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) = \sqrt{2} e^{-i\frac{\pi}{3}}$$

$$\text{Now } \alpha^{14} + \beta^{14} = \left(\sqrt{2} e^{i\frac{\pi}{3}} \right)^{14} + \left(\sqrt{2} e^{-i\frac{\pi}{3}} \right)^{14}$$

$$\Rightarrow \alpha^{14} + \beta^{14} = (\sqrt{2})^{14} \left(e^{i\frac{14\pi}{3}} + e^{-i\frac{14\pi}{3}} \right)$$

$$\Rightarrow \alpha^{14} + \beta^{14} = (\sqrt{2})^{14} \left(\cos \frac{14\pi}{3} + i \sin \frac{14\pi}{3} + \cos \frac{14\pi}{3} - i \sin \frac{14\pi}{3} \right)$$

$$\Rightarrow \alpha^{14} + \beta^{14} = 2^7 \left(2 \cos \frac{14\pi}{3} \right)$$

$$\Rightarrow \alpha^{14} + \beta^{14} = 2^8 \left(\cos \left(4\pi + \frac{2\pi}{3} \right) \right)$$

$$\Rightarrow \alpha^{14} + \beta^{14} = 2^8 \left(\cos \frac{2\pi}{3} \right)$$

$$\Rightarrow \alpha^{14} + \beta^{14} = -2^8 \times \frac{1}{2} = -128$$

8. (4)

Given,

$$x|x| - 5|x + 2| + 6 = 0$$

Now taking case (1) when $x < -2$ we get,

$$-x^2 + 5x + 10 + 6 = 0$$

$$\Rightarrow x^2 - 5x - 16 = 0$$

$\Rightarrow x = \frac{5 \pm \sqrt{89}}{2}$, ignoring $x = \frac{5 + \sqrt{89}}{2}$ as $x < -2$, so one solution is possible here,

Now taking case (2) when $-2 \leq x < 0$ we get,

$$-x^2 - 5x - 10 + 6 = 0$$

$$\Rightarrow x^2 + 5x + 4 = 0$$

$\Rightarrow x = -1$, ignoring $x = -4$ as $-2 \leq x < 0$

Now taking case (3), when $x \geq 0$ we get,

$$x^2 - 5x - 10 + 6 = 0$$

$$\Rightarrow x^2 - 5x - 4 = 0$$

$\Rightarrow x = \frac{5 \pm \sqrt{41}}{2}$, ignoring $x = \frac{5 - \sqrt{41}}{2}$ as $x \geq 0$

So, from all cases we get total 3 solutions.